

Assembly Manual

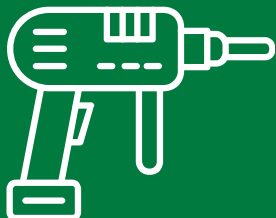
How to
Re - Move

1

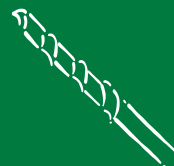


3D Printer - Bambu Lab

2



Drill + 8.5 mm drill bit

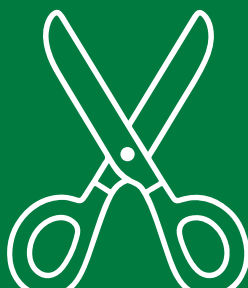


3



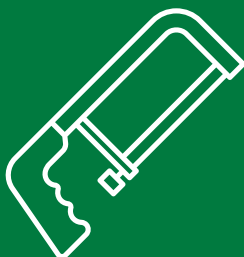
Lighter

4



Scissors

5

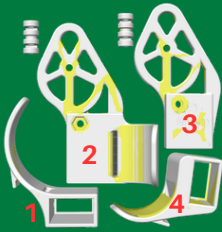


Hacksaw

Tools

Use tools at your own risk and follow proper safety precautions.

1



Download the plate (right or left leg) and print it using a Bambu Lab 3D printer.

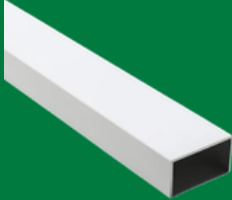
If you don't own a Bambu Lab printer, you can print it at your local Fab Lab:

<https://www.fablabs.io/>

3D Printed

The pieces are numbered for a better assembly

2



Cut 2 pieces (235 mm each) using a hacksaw.

25x45 mm aluminium profile

3



Cut 2 pieces, about 400 mm each. Then melt the ends with a lighter to prevent fraying

Rope

4



Springs x4

5

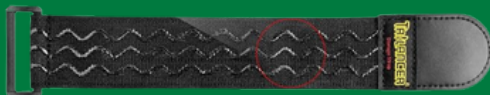


M8 bar: cut with the hacksaw

- 2x 55 mm
- 2x 35 mm
- 1x 30 mm

M8: bar + 10 nuts

6



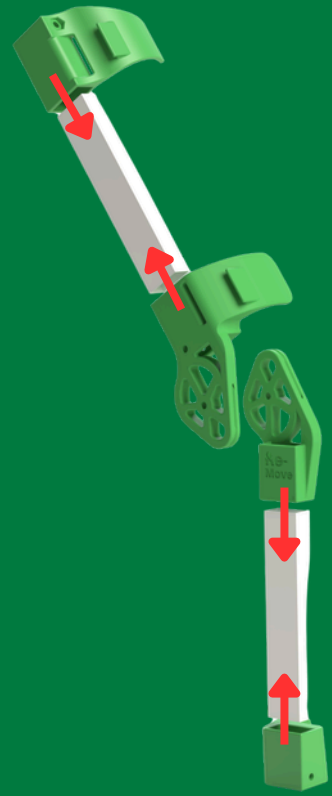
Velcro Straps

Parts

Files and links for parts are available in **GitHub**:

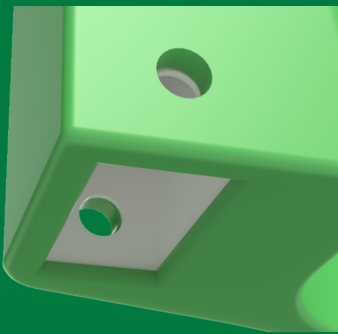
<https://github.com/Marcell-Hansmann/Re-Move-Exoskeleton>

Step 1 Assemble pieces 1, 2, 3 and 4 with aluminium bars



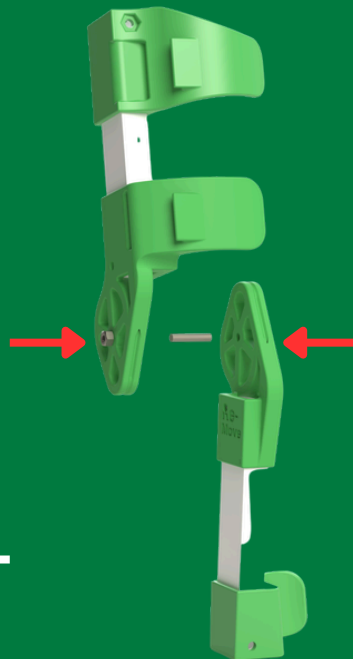
Step 2 Drill a 8.5mm hole in the aluminium profiles Use the holes in the 3D-printed part as a guide

8 holes in total, 4 for each bar



Use M8 bar size 30mm and 2 nuts

Step 3 Join the two parts

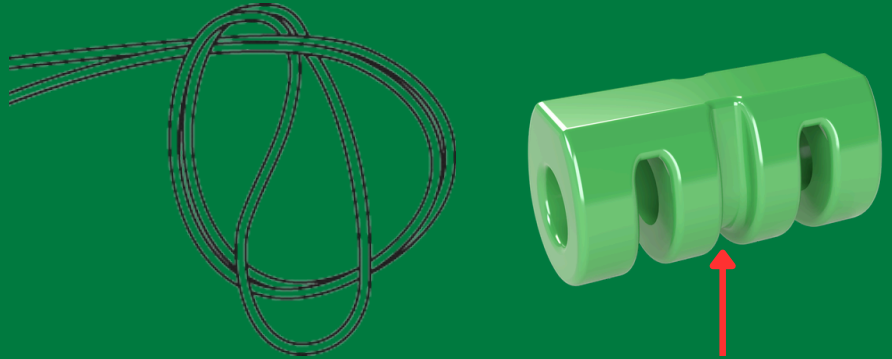


Structural assembly

Step 1

Tie an overhand loop at the end of each rope and attach it to the center of each spring unifier

1 x Rope 400 mm
1 x 3D Printed Spring unifier

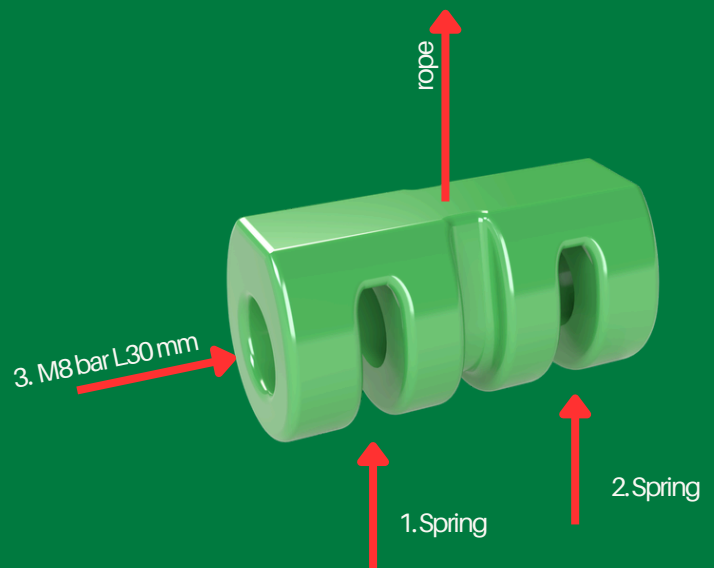


Step 2

Assemble each set of 2 springs with the spring unifiers

1 x M8 bar L30 mm
2 x Spring

Follow the correct order, as shown in the image



Step 3

Repeat the process with the second spring unifier

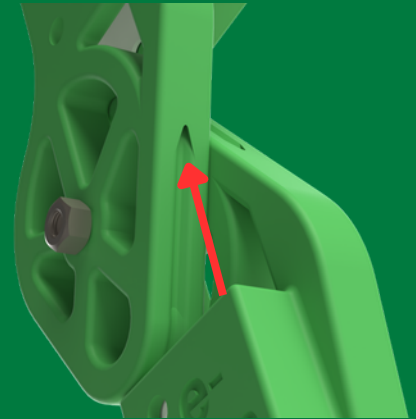
Mechanism assembly

Step 4 Assemble the spring sub-assembly inside the aluminium profiles

1 x M8 bar L55 mm
2 x M8 nut

Use the **55 mm** M8 bars to fix the springs from the end

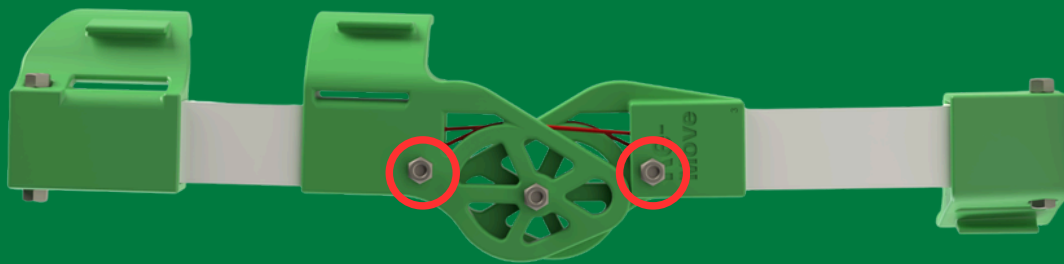
Pass the other end of the rope through the hole of the opposite CAM



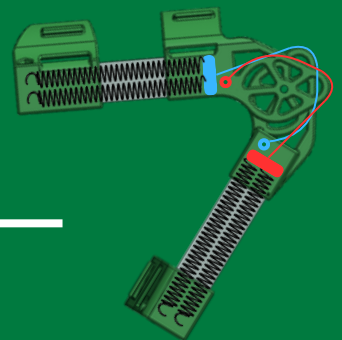
Step 5 In the resting position (leg straight), knot another overhand loop at the marked position (○)

1 x M8 bar L35 mm
2 x M8 nut

Then fix it by passing the **35mm** bar through with the remaining nuts
Cut the excess rope and burn the ends again



Step 6 Repeat the process for the opposite side. The result should match the diagram:



Mechanism assembly

Step 1 Add the Velcro straps to each part

3x Velcro straps

Pass the Velcro strap through the hole, with the Velcro side facing the 3D-printed material.
The result should be as shown in the images below
Repeat the process for all 3 of the parts



Step 2 Attach to your leg

Secure each part around your leg using the Velcro straps.
Adjust the tightness of the Velcro straps
Make sure to align your knee axis with the exoskeleton axis properly

Warning!

This device is for experimental purposes only and is not medically certified.



Attach to your leg